

Early and Later EEG Synchronization During 40-Hz Auditory Stimulation

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One strong responder's session (sub-32): 40-Hz phase-locking value in each 20-s window, 9 complete cycles

■ sound on ■ silence

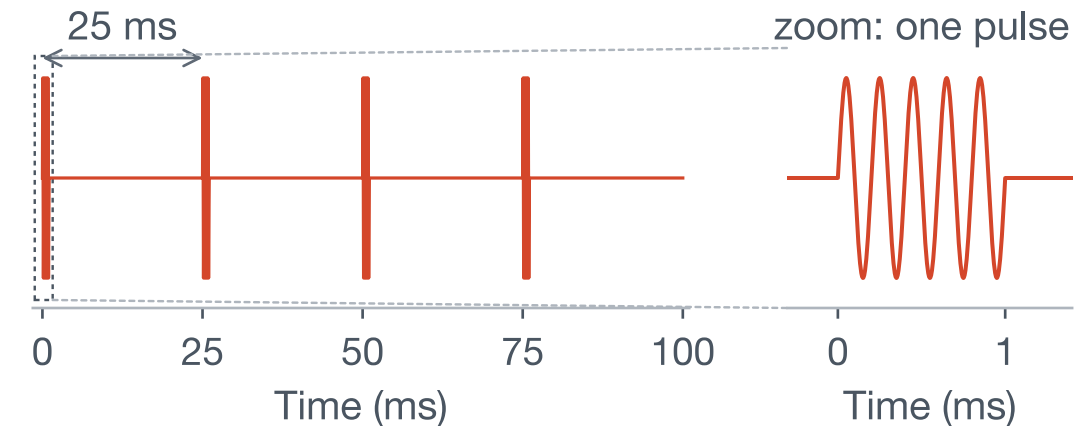
An early EEG response can only help tailor 40-Hz stimulation if it tracks the later response

In mice, 40-Hz light and sound reduced Alzheimer's-like pathology.^{1,2} Replication has been mixed,³ and benefit in patients is not established.^{4,5}

Responses also differ between people: not every brain follows 40-Hz sound.⁶ Tailoring stimulation to a person would need a response measure taken **early** in the session.

The question: **within one session, does a person's EEG response in the first two minutes track their response a few minutes later?**

The stimulus

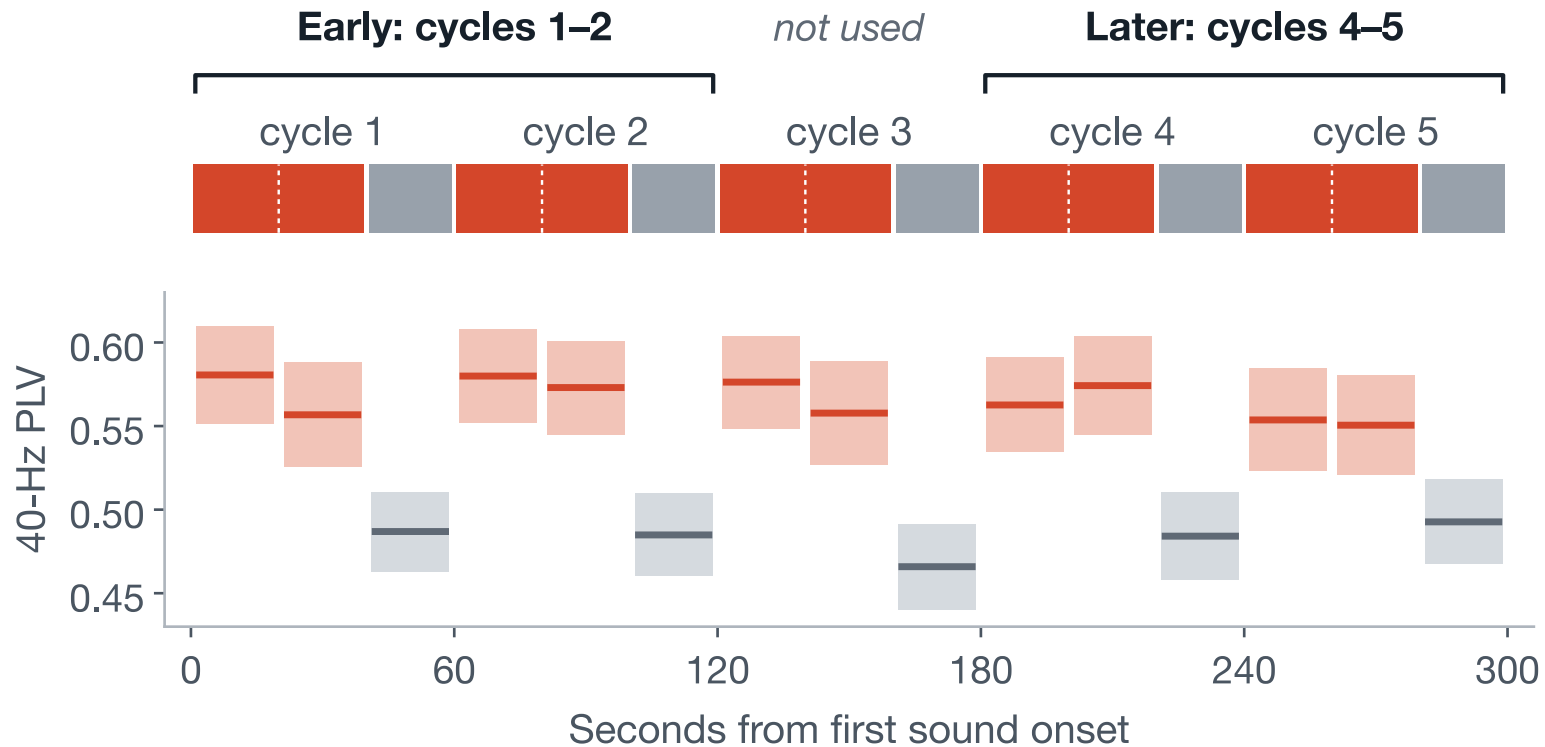


A 1-ms, 5-kHz tone pip every 25 ms, so 40 pulses per second. Played from two loudspeakers, eyes open.⁶

¹Iaccarino et al., *Nature* 2016 ²Martorell et al., *Cell* 2019 ³Soula et al., *Nat Neurosci* 2023

⁴Chan et al., *PLOS ONE* 2022 ⁵Hajós et al., *Front Neurol* 2024 ⁶Lahijanian et al., *Sci Rep* 2024

35 people heard 40-s blocks of 40-Hz sound, each followed by 20 s of silence



■ sound ■ silence

Bottom: group-mean 40-Hz phase-locking value (PLV) in each 20-s window, $n = 35$; bands ± 1 SE across people. Each 40-s sound block gives two windows.

Public EEG from a memory clinic in Tehran (OpenNeuro ds005048)¹

10 cognitively normal, 6 with mild cognitive impairment, 17 with Alzheimer's, 2 unclassified

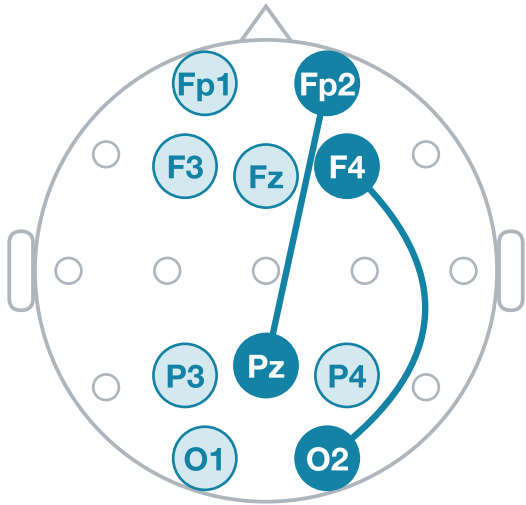
5 ($n = 8$) or 9 ($n = 27$) complete cycles per session

$$\langle \text{PLV}^{\text{sound}} - \text{PLV}^{\text{silence}} \rangle = +0.077$$

Whole session, mean of 35 people, 95% CI [0.045, 0.112]; higher with sound in 27 of 35.

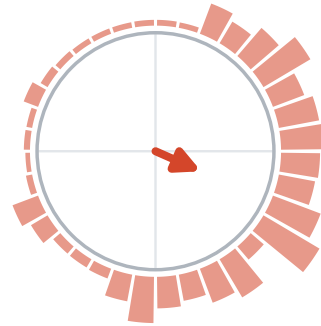
¹Lahijanian, Aghajan & Vahabi, *Sci Rep* 14:13153 (2024); data doi:10.18112/openneuro.ds005048.v1.0.1. 19-channel EEG at 250 Hz, cleaned by the dataset authors. Sessions had 6 or 10 sound blocks; the last silence is cut short.

Phase-locking value measures how steady the timing is between two 40-Hz signals



25 bipolar signals (frontal minus posterior), forming 200 pairs that share no electrode. Lines: the example pair, Fp2–Pz and F4–O2.

Sound on

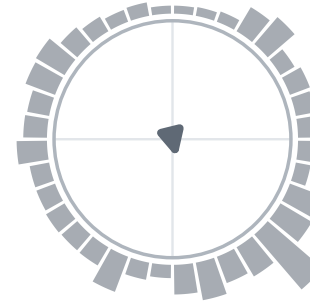


$$PLV_{ab} = 0.42$$

Median pair of sub-01 (largest early contrast), cycle 1. Bars: where the phase difference $\Delta\phi_{ab}$ fell. Arrow length is the PLV (circle radius = 1).

$$PLV_{ab} = \left| \frac{1}{N} \sum_{n=1}^N e^{i\Delta\phi_{ab}(n)} \right|$$

Silence



$$PLV_{ab} = 0.16$$

$N = 5000$ samples per window. Reported PLV: the mean over the 200 pairs.

Contrast for cycle k

$$C_k = \overline{PLV}_k^{\text{sound}} - PLV_k^{\text{silence}}$$

Sound: mean of the block's two 20-s windows. Silence: the 20 s right after.

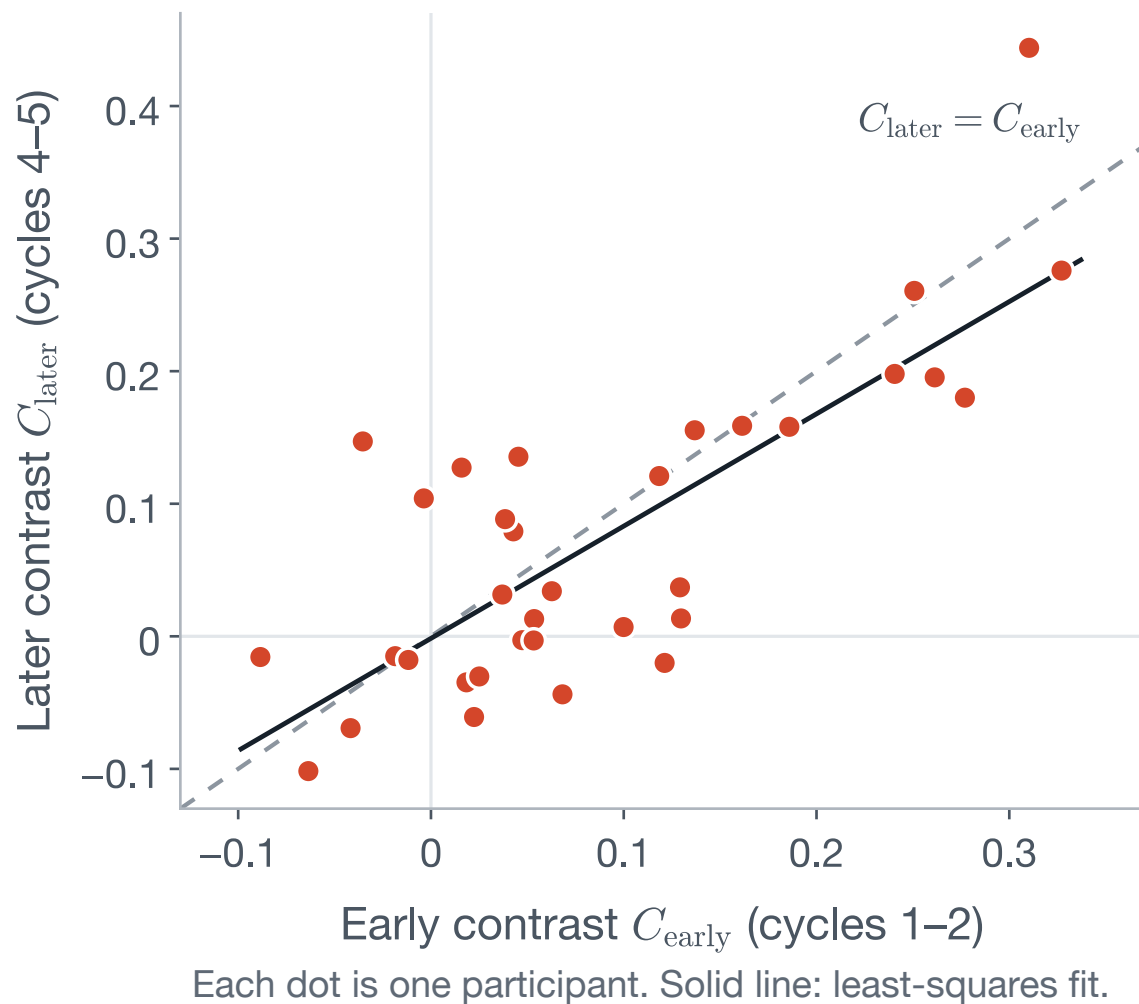
$$C_{\text{early}} = \frac{1}{2} (C_1 + C_2)$$

$$C_{\text{later}} = \frac{1}{2} (C_4 + C_5)$$

Why subtract silence?

At every frequency, raw PLV in block 1 correlated with the last three blocks ($0.87 \leq r \leq 0.93$), even where sound barely changes it. It mostly reflects the person.

People with a larger early contrast had a larger later contrast



Early vs. later contrast at 40 Hz

$$r = 0.780$$

95% CI [0.573, 0.889], $n = 35$

Pearson; participant bootstrap, 50,000 resamples

Ranks agree too (Spearman $\rho = 0.688$), and so do the values themselves (Lin's $\rho_c = 0.771$).

No clear shift in the group mean:

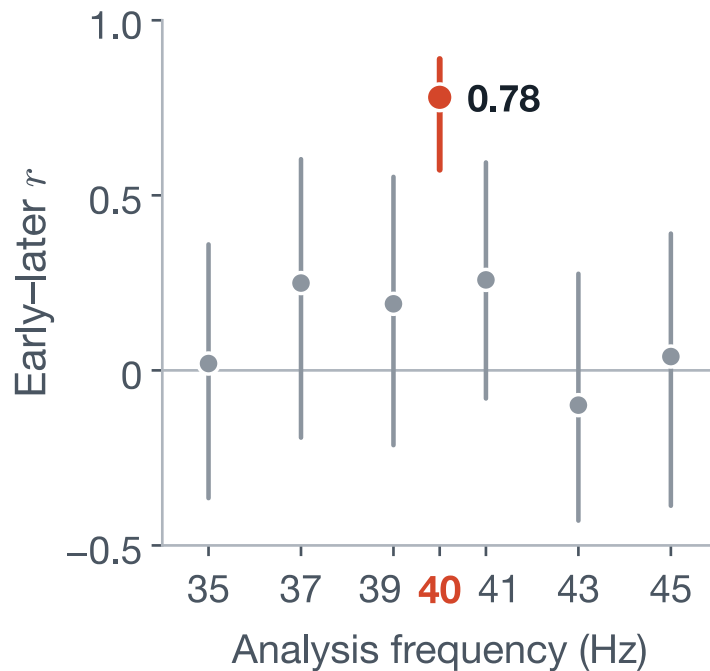
$$\langle C_{\text{later}} - C_{\text{early}} \rangle = -0.015 \quad (p = 0.25).$$

Individuals still moved: 95% limits of agreement $[-0.16, +0.13]$.

The link was strongest at 40 Hz and held across measures and later windows

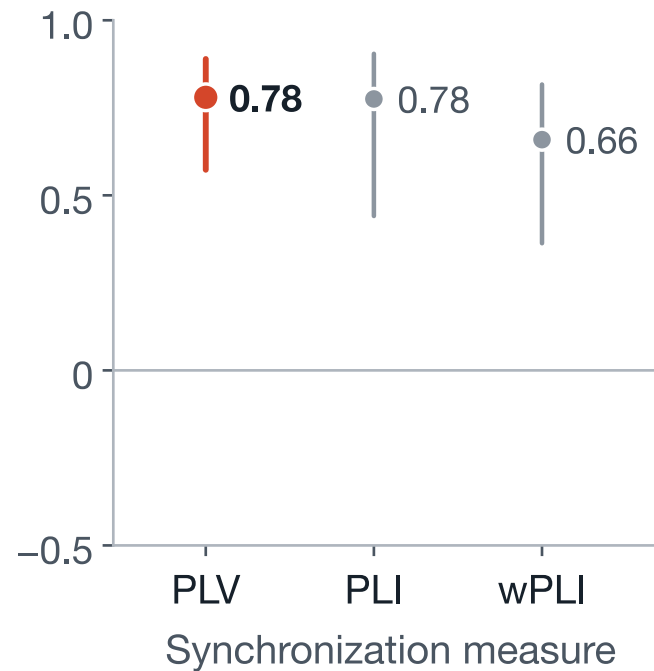
Dots: early-later Pearson r . Whiskers: 95% participant-bootstrap CIs. The early window is always cycles 1–2.

1. Strongest at 40 Hz?



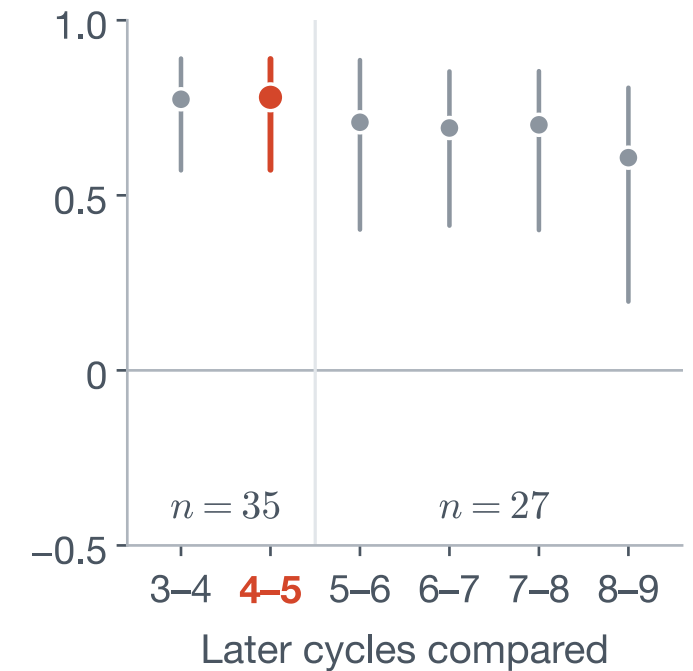
Six neighbors: $-0.10 \leq r \leq 0.26$.
Smallest gap to 40 Hz: $\Delta r = 0.52$,
95% CI [0.08, 0.65].

2. Only with PLV?



PLI and wPLI discount phase differences near 0 or $\pm\pi$, the mark of one source reaching both signals.

3. Were cycles 4–5 a lucky pick?



They gave the highest r ; the rest gave $0.61 \leq r \leq 0.77$, all $p < 0.05$ after Bonferroni correction.

An early, within-session EEG measure worth testing prospectively

What I found

The contrast in the **first two minutes** tracked the contrast 3 to 5 minutes in ($r = 0.78$), most strongly at 40 Hz.

What limits it

Retrospective: cycles 4–5 chosen after seeing the data.

Offline: cleaned using whole recordings.

No sham: silence always followed sound.

Scalp level: a shared response or artifact not ruled out.

Still untested

Repeatability across sessions, use for guiding stimulation, and any link to clinical benefit.

Next

A preregistered, real-time test with a sham or counterbalanced order and repeat sessions.

Data: Lahijanlian, Aghajan & Vahabi, OpenNeuro ds005048 (CC0). Thank you to the participants and the dataset authors.